

Simultaneous Multitone SQR in Dispersive cQED: When One Common Waveform Can and Cannot Drive Multiple Branches

March 19, 2026

1 Motivation

The earlier SQR waveform study showed that strict logical performance can fail even when relaxed branchwise metrics look strong. This follow-up asks a harder question: can one common multitone waveform implement *multiple* selective qubit rotations at the same time? If the answer is no for the simplest family, we also want to know whether the failure is primarily:

1. a removable cavity block-phase profile,
2. a simple pulse-calibration problem,
3. or a deeper limitation of the common Gaussian multitone ansatz.

The key scientific caution in this study is that fidelity alone can be misleading for small target angles. If a pulse acts almost like identity, then it can still score well against a small-angle target simply because the target itself is close to identity. We therefore benchmark both fidelity and the actually executed target-branch rotation angle.

2 System & Methods

We model a dispersive transmon-cavity system with

$$\omega_q = 2\pi \times 6.150 \text{ GHz}, \quad \omega_c = 2\pi \times 5.241 \text{ GHz}, \quad \alpha = 2\pi \times (-255 \text{ MHz}), \quad \chi = 2\pi \times (-2.84 \text{ MHz}).$$

The logical cavity window is $n = 0, 1, 2, 3$. The main targeted-subspace study uses $n_{\text{tr}} = 2$ and $n_{\text{cav}} = 6$, because the current ‘cqed_sim’ targeted-subspace multitone workflow evaluates conditioned qubit metrics through a two-level reduced qubit state. Representative waveforms are then replayed on $n_{\text{tr}} = 3$ to estimate $|f\rangle$ leakage.

We study four simultaneous target sets:

$$\{0, 1\}, \quad \{0, 2\}, \quad \{0, 1, 2\}, \quad \{0, 1, 2, 3\},$$

with target rotation angles

$$\theta/\pi \in \{1/8, 1/4, 1/2, 1\},$$

always around the X axis. The duration scan uses

$$\chi T/2\pi \in \{1, 2, 3, 5\}.$$

For each point we compute:

- reduced conditioned multitone fidelity,
- strict logical targeted-subspace fidelity,
- best-fit cavity block-phase compiled fidelity,
- state-transfer fidelity,
- same-block population preservation,
- best-fit cavity block phases,
- and the mean simulated target-branch Bloch angle $\bar{\theta}_{\text{sim}}$.

We additionally test three correction ideas on a representative hard case (pair target $\{0,1\}$, $\theta = \pi$, $\chi T/2\pi = 3$):

1. common target-tone amplitude correction $\Delta\lambda$,
2. multistart amplitude/phase/detuning optimization,
3. naive repeated-step compilation using eight compiled $\pi/8$ steps.

Validation includes sanity checks, convergence checks in dt and n_{cav} , and qutrit replay spot-checks.

3 Results

3.1 Baseline Fidelity Maps Are Misleading Without an Angle-Response Diagnostic

Figure 1 shows the reduced conditioned fidelity over the full scan. Taken at face value, the plots could suggest that simultaneous multitone SQR is good for small angles and only breaks down for large angles.

The crucial correction is Figure 2. At the representative $\chi T/2\pi = 3$ slice, the mean simulated target angle is tiny:

$$\bar{\theta}_{\text{sim}}/\pi \sim 10^{-5}\text{--}10^{-4},$$

even when the requested target is $\theta = \pi$. For the pair-adjacent case $\{0,1\}$, the target-angle response ratios are

$$\bar{\theta}_{\text{sim}}/\theta \approx 2.23 \times 10^{-4}, 2.23 \times 10^{-4}, 2.22 \times 10^{-4}, 2.20 \times 10^{-4}$$

for $\theta/\pi = 1/8, 1/4, 1/2, 1$, respectively. The other target sets show the same behavior at the 10^{-4} level.

This changes the interpretation completely: the common Gaussian simultaneous multitone family is not a successful multi-SQR pulse that merely struggles at large angle. It is, to leading order, an almost-identity pulse on every block. The apparently strong small-angle performance arises because $R_X(\theta)$ is itself close to identity for small θ .

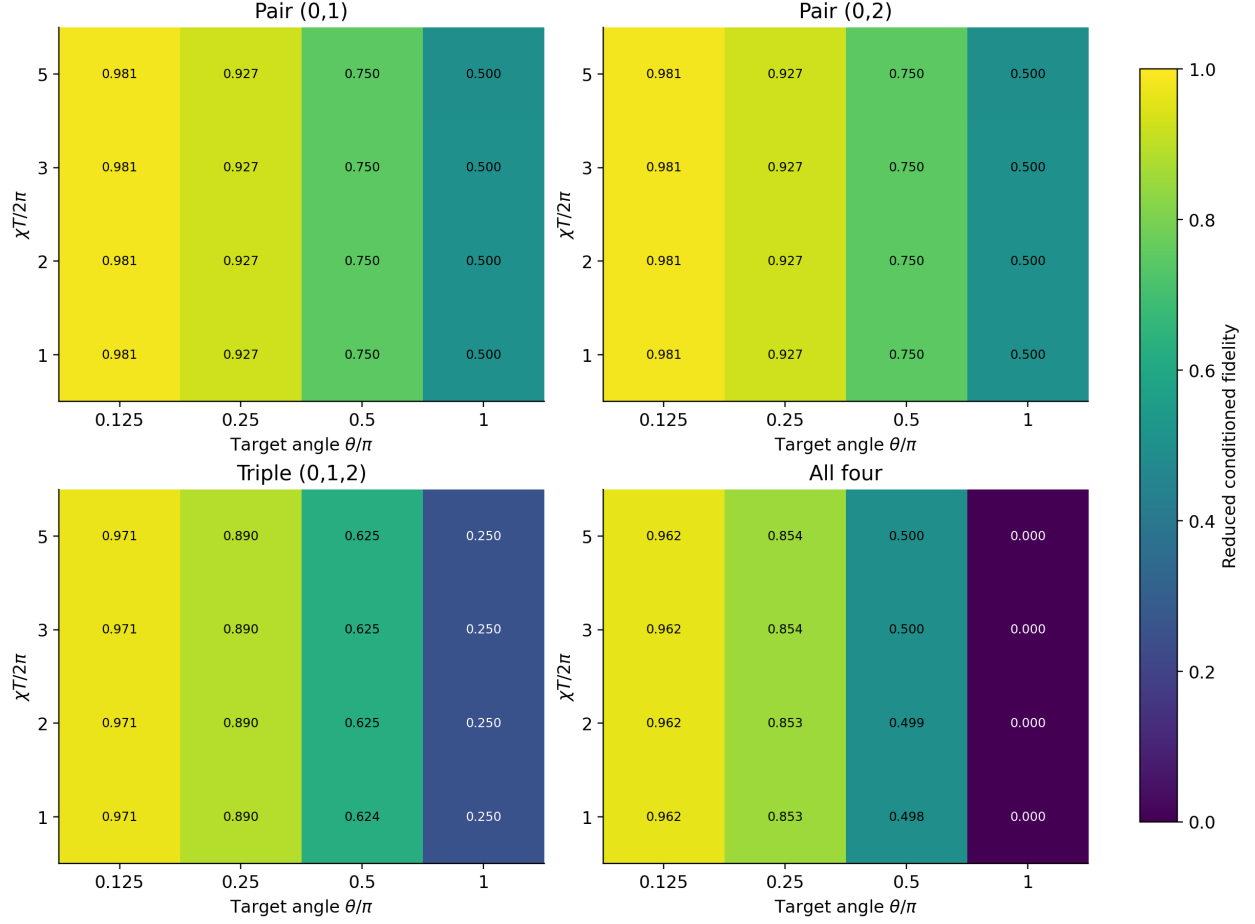


Figure 1: Reduced conditioned fidelity for simultaneous multitone targets. Fidelity decreases with requested angle, but this plot alone does not tell us whether the pulse is really implementing the requested rotation or merely staying close to identity.

3.2 Strict Logical Validation Shows Block Preservation Without Useful Rotation

Figure 3 gives the strict targeted-subspace logical fidelity. The scan is nearly independent of $\chi T/2\pi$ over the explored range: changing duration from 1 to 5 improves the representative pair-target $\theta = \pi$ strict fidelity only from 0.2476 to 0.2483.

Representative $\chi T/2\pi = 3$ strict fidelities are:

Target set	$\theta/\pi = 1/8$	$1/4$	$1/2$	1
$\{0, 1\}$	0.9759	0.9205	0.7242	0.2477
$\{0, 2\}$	0.9759	0.9205	0.7244	0.2479
$\{0, 1, 2\}$	0.9664	0.8843	0.6051	0.0616
$\{0, 1, 2, 3\}$	0.9570	0.8490	0.4970	3.8×10^{-7}

These values look respectable for small θ , but Figure 2 shows that they do *not* correspond to correct simultaneous rotations.

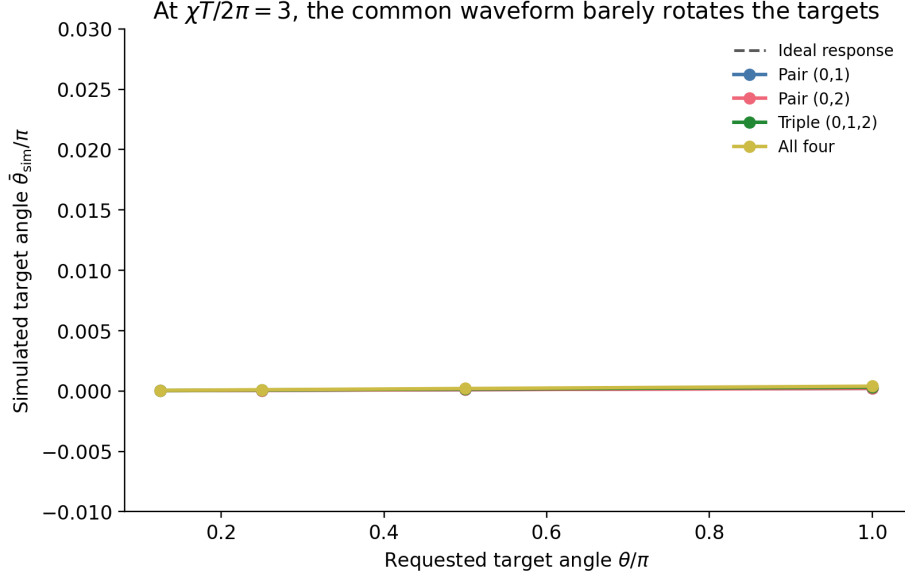


Figure 2: Actual target-angle response at $\chi T/2\pi = 3$. The ideal diagonal line would mean the waveform implements the requested rotation. The simulated response instead stays near zero for every simultaneous target set, so the nominal “good” small-angle fidelity is mostly trivial identity overlap.

At the same time, the pulse preserves cavity blocks extremely well: the minimum same-block population mean over the whole scan is

$$0.999998207.$$

Representative qutrit replay spot-checks also show tiny transmon leakage, with a maximum observed $|f\rangle$ population of only

$$7.08 \times 10^{-9}.$$

So the dominant failure is neither block mixing nor obvious qutrit leakage. The waveform simply fails to generate the requested qubit rotations.

3.3 Cavity Block-Phase Compilation Is Real but Secondary

The best-fit cavity block-phase profile is extremely regular. At $\chi T/2\pi = 3$, a representative best-fit phase vector is approximately

$$\phi_n \approx (0, -0.03267, -0.06535, -0.09802) \text{ rad},$$

which is nearly linear in n . However, the compiled gain remains tiny: the largest best-fit cavity-compilation gain anywhere in the scan is only

$$\Delta F_{\text{comp}}^{\text{max}} = 1.47 \times 10^{-3}.$$

So a cavity phase exists, but it is not the main obstacle. The main obstacle is that the common waveform does not perform the intended block-local rotation in the first place.

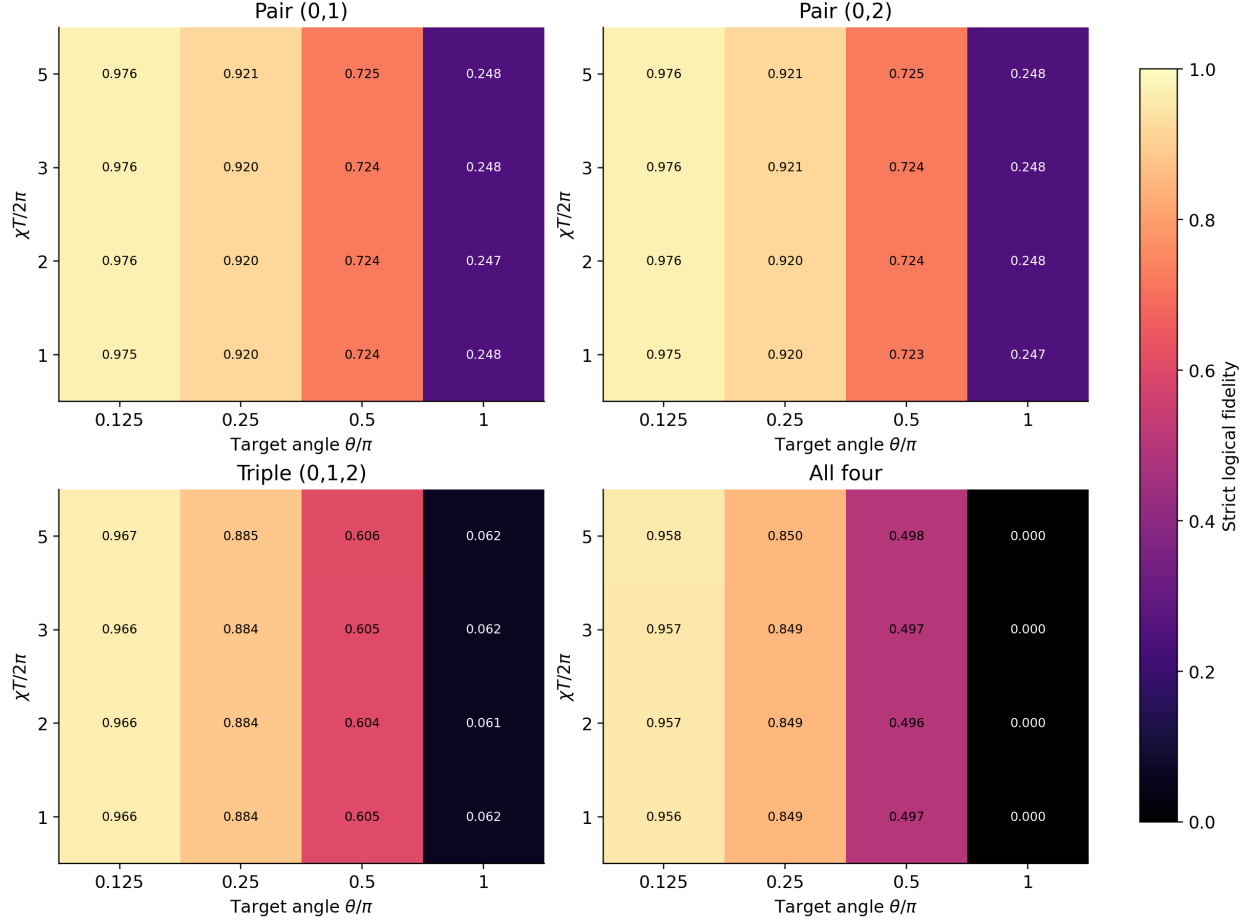


Figure 3: Strict logical targeted-subspace fidelity. These values are lower than the reduced conditioned metrics, but the same qualitative warning remains: small-angle numbers are dominated by identity overlap, while large-angle simultaneous targets fail strongly.

3.4 Simple Pulse Corrections Do Not Rescue the Hard Regime

Figure 5 summarizes the correction studies for the hard pair-target π -gate at $\chi T/2\pi = 3$.

The explicit results are:

- Sweeping a common target-tone amplitude correction $\Delta\lambda \in [-1.5, 3.0]$ changes the reduced mean fidelity only from 0.50000000 to 0.50000007.
- Three multistart waveform-side optimizations over amplitude, phase, and detuning reach a best reduced fidelity of only 0.50000040. The best corresponding strict logical fidelity is 0.231997, actually below the raw baseline strict fidelity 0.247713.
- Thus, simple pulse correction within this ansatz does not rescue the simultaneous π -gate regime.

3.5 Naive Repeated Small-Angle Compilation Also Fails

One possible fallback idea is to use many high-fidelity small-angle primitives instead of one direct π pulse. We tested this by composing the best-fit cavity-compiled restricted operator for the $\pi/8$

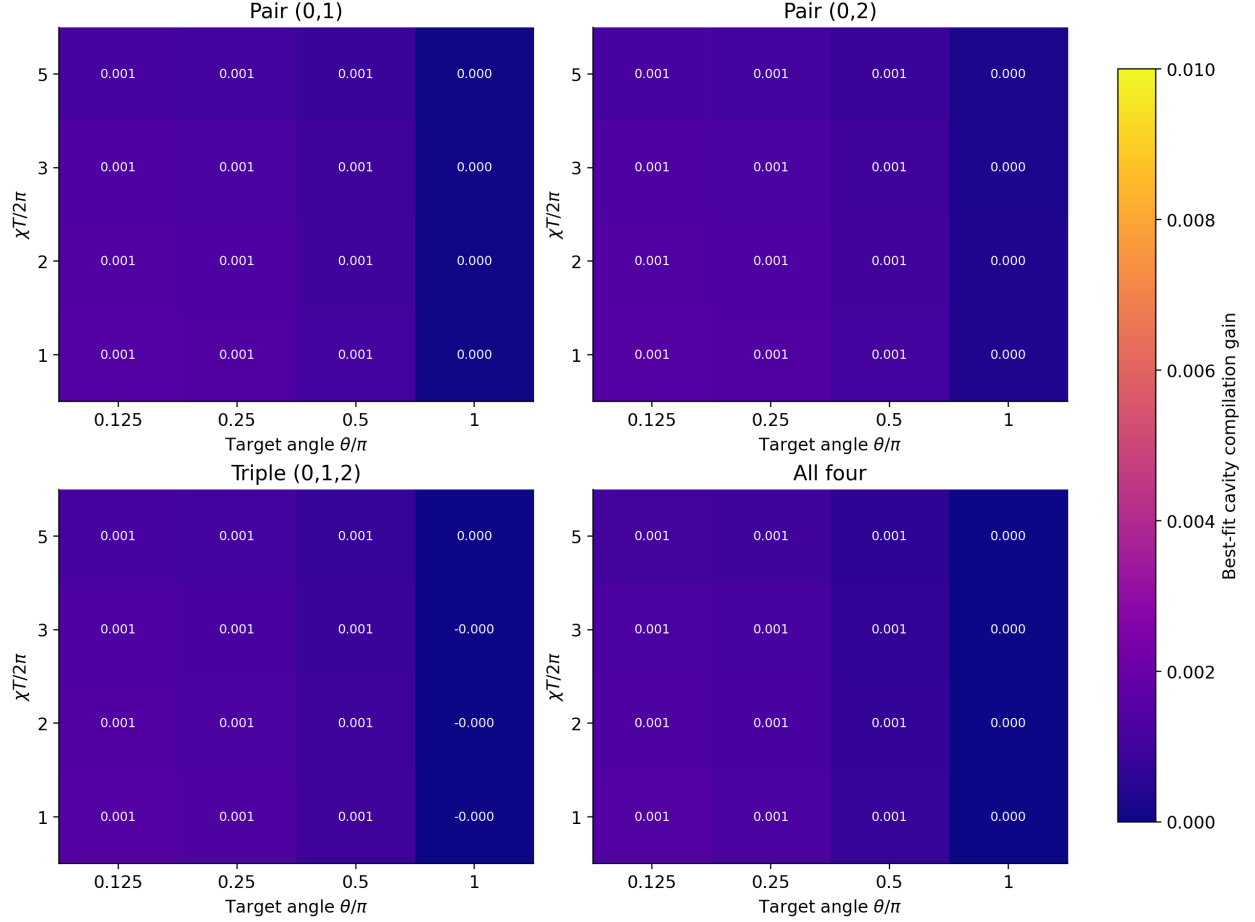


Figure 4: Best-fit cavity-only compilation gain. The induced block phase is smooth and real, but correcting it changes the conclusion only at the 10^{-3} level. This is not primarily a cavity-phase-compilation problem.

step eight times and comparing the resulting logical operator with the desired direct π target.

The resulting operator-level fidelities are:

$$F_{\text{seg}}^{\{0,1\}} = 0.1546, \quad F_{\text{seg}}^{\{0,1,2\}} = 0.0311,$$

compared with direct- π strict fidelities 0.2477 and 0.0616, respectively. Naive repeated small steps therefore do not solve the problem.

4 Discussion

The central conclusion is that the common Gaussian simultaneous multitone family studied here is *not* a practical simultaneous multi-SQR primitive. The strongest evidence is not the fidelity maps alone but the explicit target-angle response: the waveform leaves every target branch almost unrotated, with only a tiny $O(10^{-4})$ fractional response to the requested angle.

This also explains several otherwise puzzling features:

- Pair-adjacent and pair-separated targets perform almost identically, so nearest-neighbor branch crowding is not the main issue.

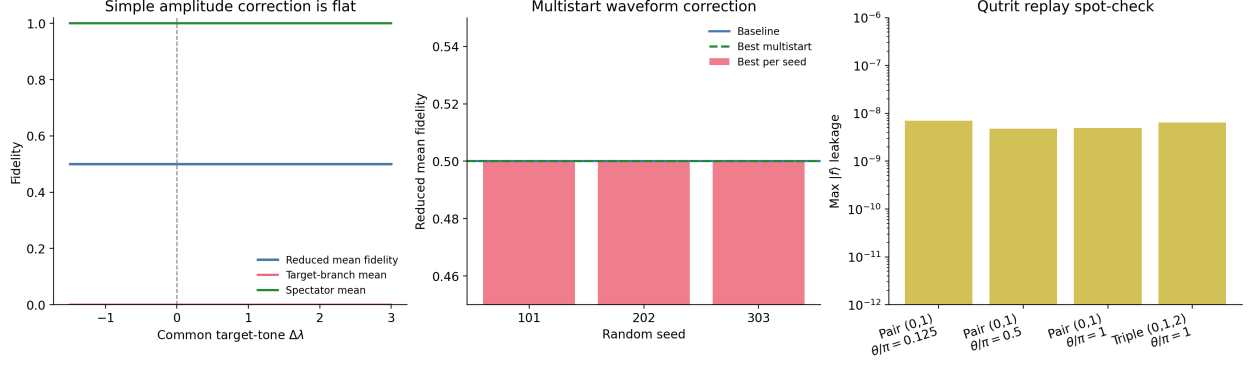


Figure 5: Correction checks for the representative hard case. Left: sweeping a common target-tone amplitude correction is essentially flat. Middle: broad multistart amplitude/phase/detuning correction remains stuck near the same reduced fidelity. Right: qutrit replay spot-checks confirm the failure is not driven by large $|f\rangle$ leakage.

- Increasing $\chi T/2\pi$ over the scanned range barely changes the outcome, so duration alone does not fix the ansatz.
- The cavity block phase is smooth and nearly linear, but correcting it helps only at the 10^{-3} level because the principal defect is missing rotation, not missing phase compilation.
- Tiny qutrit leakage and excellent same-block preservation show that the pulse is coherent and block-preserving; it is just the wrong coherent action.

From an experiment-facing control perspective, the lesson is very specific: before trusting any simultaneous-target fidelity metric, one must calibrate the actual target-branch angle response. In this study, the nominal parameter θ in the simultaneous common-waveform builder is not a reliable proxy for the executed branch rotation.

The likely next steps are therefore not more minor corrections to this exact family, but richer waveform classes:

1. segmented simultaneous pulses with independently shaped subwindows,
2. explicitly time-dependent tone phases or chirps,
3. or direct optimal control constrained by hardware bandwidth.

Only after such a waveform already produces the correct block-local rotations does cavity phase compilation become a meaningful secondary cleanup layer.

5 Conclusion

The study answers the original question in a fairly decisive way.

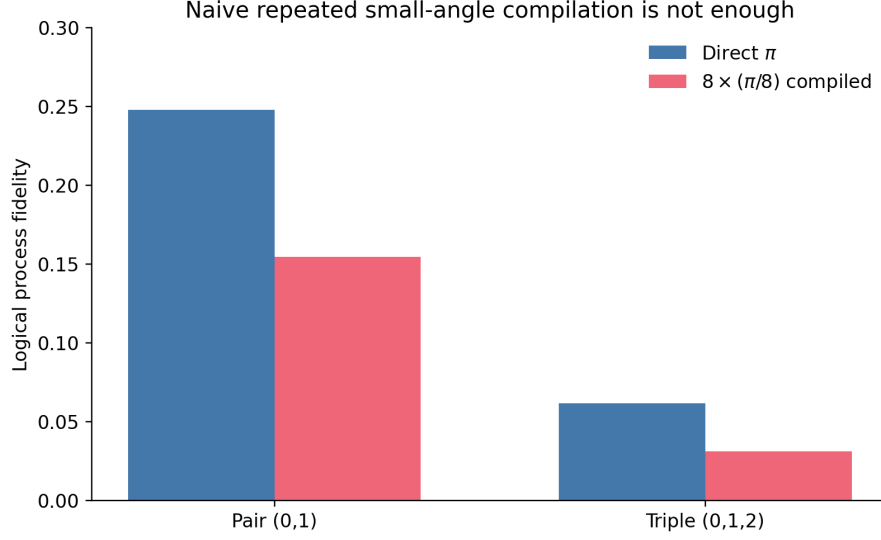


Figure 6: Naive repeated-step compilation is not enough. Eight compiled $\pi/8$ steps do not outperform the already-bad direct π gate for either the pair or triple target sets.

Primitive idea	Verdict	Main issue	Recommendation
Direct common Gaussian simultaneous multitone SQR	Not viable	Target branches barely rotate; fidelity can look good only because small-angle targets are near identity	Do not deploy as a simultaneous multi-SQR gate
Direct pulse + cavity block-phase compilation	Not enough	Compilation gain is only $O(10^{-3})$	Use only after a better waveform family is found
Simple amplitude correction	Not enough	Response is essentially flat in the hard regime	Not worth pursuing alone
Broad multistart amplitude/phase/detuning correction	Not enough	No meaningful gain over baseline	Treat as evidence of ansatz limitation, not just poor initialization
Naive repeated small-angle compilation	Not enough	Composition of weak near-identity steps does not build the desired π gate	Do not use without a richer inter-step compilation strategy

The best immediate recommendation is:

Do *not* use the common Gaussian simultaneous multitone family as an experiment-facing simultaneous multi-SQR primitive. Move instead to richer segmented or optimal-control waveforms, and evaluate them with explicit target-angle response diagnostics in addition to fidelity.

The most important unresolved bottleneck is not cavity phase. It is the absence of a waveform family that can produce the desired simultaneous block-local rotations at all.